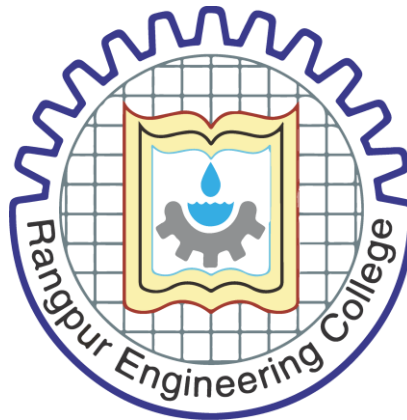


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**WeavePDF**

A Project Proposal  
Course code: CSE 4192  
Course Title: Thesis/ Project (Part I)

**Submitted By**

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# WeavePDF

All-in-One PDF & Productivity Web Platform

## Project Proposal

|                       |                                                      |
|-----------------------|------------------------------------------------------|
| <b>Project Title</b>  | WeavePDF: All-in-One PDF & Productivity Web Platform |
| <b>Project Type</b>   | Full-Stack Web Application (With NextJS & MongoDB)   |
| <b>Target Users</b>   | Students, Professionals, Businesses, Remote Teams    |
| <b>Proposed Stack</b> | Next.js, React, Node.js, MongoDB, JWT, Tailwind CSS  |

## 1. Introduction

In today's digital-first world, professionals, students, and businesses constantly deal with documents editing, converting, sharing, and managing them across different platforms. The fragmented ecosystem of single-purpose tools forces users to juggle between multiple websites and applications just to accomplish basic document tasks. There is a growing need for a unified, intelligent, and secure platform that centralizes these workflows.

WeavePDF is proposed as a comprehensive, browser-based platform that brings together powerful PDF management, document conversion, important tools, real-time collaboration, and productivity features under one roof. By integrating seamlessly designed modules from PDF editing and speech-to-text conversion to a built-in messenger/WhatsApp, quiz generator, and resume builder - WeavePDF aims to become the single destination for all document-related needs. And This will provide separate features for premium users.

The platform will also feature a secure digital payment gateway, enabling users to access premium tools and services with confidence. WeavePDF is designed to serve a wide audience, from individual users and students to enterprise teams, offering an intuitive dashboard for managing activity, files, and users efficiently.

## 2. Problem Statement

At present, users who need to work with PDF documents and related productivity tools are forced to navigate across many separate platforms. A student may use one website to convert a Word document to PDF, another to add a watermark, a third to generate a quiz from study material, and yet another tool for collaboration. Each switch causes friction, wastes time, and introduces security risks when sensitive documents are uploaded to unfamiliar third-party services.

The key problems this project addresses are:

- Lack of a unified platform combining PDF tools, collaboration, and productivity features.
- Privacy concerns when uploading sensitive documents to multiple third-party services.
- No seamless payment integration for accessing premium document management services.
- Difficulty in building CVs, resumes, or quizzes without switching between multiple tools.
- Limited accessibility users without strong technical backgrounds find it hard to manage complex PDF workflows.
- No built-in communication or collaboration layer within document management platforms.

### 3. Objectives

The primary objectives of WeavePDF are:

- To develop a fully functional, responsive web application for PDF management and document productivity.
- To integrate important features including speech-to-text, quiz generation from PDFs, and PDF-to-audiobook conversion.
- To implement a secure digital payment gateway for premium feature access and service monetization.
- To provide real-time collaboration through an integrated messenger chat system.
- To offer document personalization tools including signature drawing, watermarking, and page numbering.
- To build resume and CV builders that help users create professional documents without external tools.
- To deliver an admin dashboard with insights, user management, and download tracking.
- To ensure a secure, scalable, and user-friendly experience using modern full-stack technologies.
- Unlock advanced tools and exclusive functionality through premium subscription plans.

### 4. Key Features

#### 4.1 Digital Payment Gateway

A secure and smooth payment processing system will be integrated to support premium tool access. Users will be able to subscribe to plans or make one-time payments using trusted payment providers, with transaction history and billing management available from the dashboard.

#### 4.2 Merge & Split PDF

Users can combine multiple PDF files into a single document or divide a large PDF into smaller, more manageable parts. This module ensures document organization remains easy and efficient for both personal and professional use.

### **4.3 Edit PDF**

A rich PDF editor will allow users to add text, highlight sections, insert images, apply watermarks, rotate pages, and draw digital signatures directly within the browser without the need for any desktop software.

### **4.4 Word to PDF & Markdown to PDF**

Seamless document conversion from Microsoft Word (.docx) and Markdown (.md) formats to PDF, preserving all formatting, styles, and structure during conversion.

### **4.5 Speech to Text**

An AI-powered speech recognition module will allow users to convert spoken words into editable text in real time. This feature is particularly useful for note-taking, transcription, and hands-free content creation.

### **4.6 Image to PDF**

Users can upload one or multiple images (JPG, PNG, etc.) and convert them into a well-formatted, shareable PDF document. This is useful for digitizing notes, receipts, and photos.

### **4.7 Draw Signature & Add Watermark**

Users can draw digital signatures on a canvas and embed them into PDF documents. A watermarking tool allows branding or protection of documents with text or image overlays.

### **4.8 Add Page Numbers**

A dedicated module for adding page numbers to PDF documents will ensure that long reports, assignments, and official papers remain well-structured and easy to navigate.

### **4.9 Todo List**

A built-in task management tool will allow users to create, organize, and track their tasks directly within the platform — keeping work and document management tightly integrated.

### **4.10 Messenger/WhatsApp Chat**

A real-time messaging interface will support team collaboration, allowing users to discuss documents, share files, and coordinate tasks without leaving the platform.

### **4.11 PDF to Audiobook**

An accessibility-focused feature that converts PDF documents into spoken audio, making content more consumable for users on the move or those with reading difficulties.

### **4.12 Admin Dashboard**

A centralized dashboard for administrators to monitor platform usage, manage user accounts, track recent downloads, and view performance insights through visual analytics.

### **4.13 Resume & CV Builder**

Interactive, template-based builders that allow users to create professional resumes and CVs by filling in structured forms, with the option to export them as polished PDF documents.

#### 4.14 Quiz Generator

An AI-powered module that analyzes uploaded PDFs and automatically generates multiple-choice or short-answer quizzes ideal for students, educators, and corporate trainers.

#### 4.15 QR Code Generator

A simple and fast QR code generation tool that converts URLs, text, or document links into downloadable QR codes, useful for sharing and marketing purposes.

### 5. Technologies to be Used

| Component           | Technology                          |
|---------------------|-------------------------------------|
| Frontend Framework  | Next.js (App Router)                |
| Language            | JavaScript, TypeScript              |
| State Management    | Redux Toolkit                       |
| Styling             | Tailwind CSS, HTML5, CSS3           |
| Backend             | Node.js, Express.js                 |
| Authentication      | Json Web Token                      |
| Database            | MongoDB with Mongoose               |
| Payment Integration | Stripe or SSLCommerz                |
| PDF Processing      | Npm packages                        |
| Version Control     | Git & GitHub                        |
| API Testing         | Postman                             |
| Deployment          | Vercel (Frontend), Vercel (Backend) |

### 6. System Architecture & Workflow

WeavePDF follows a modern client-server architecture built on Next.js for both the frontend and server-side rendering layer, with a dedicated Node.js/Express.js API for backend processing. The platform uses MongoDB as the primary data store, Firebase for authentication and real-time features, and integrates external APIs for AI capabilities.

The general workflow for document processing follows these steps:

- User authenticates via JWT and lands on the home page. OR user can access without login.
- User selects a tool (e.g., Merge PDF, Speech to Text, Resume Builder).
- Files or inputs are submitted through the NextJS frontend.

- The Next.js API route or Express.js backend processes the request, calling relevant libraries or external APIs.
- The processed result (PDF, audio, text, quiz) is returned to the user for download or in-app display.
- Activity is logged to MongoDB and reflected in the admin dashboard.
- Real-time collaboration and chat will be done with messenger chat or WhatsApp chat.

## 7. Modules Breakdown

| Module                              | Description                                                                                              |
|-------------------------------------|----------------------------------------------------------------------------------------------------------|
| <b>Authentication Module</b>        | Handles user registration, login, JWT sessions, and role-based access using user credentials.            |
| <b>Payment Module</b>               | Integrates Stripe or SSLCommerz for secure subscription and one-time payment processing.                 |
| <b>PDF Processing Module</b>        | Core engine for merge, split, edit, watermark, signature, and page number operations using npm packages. |
| <b>Conversion Module</b>            | Converts Word and Markdown documents to PDF using Puppeteer and markdown-pdf.                            |
| <b>Node packages</b>                | Drives speech-to-text, quiz generation from PDFs, and PDF-to-audiobook features via npm packages.        |
| <b>Image &amp; QR Module</b>        | Handles image-to-PDF conversion and dynamic QR code generation.                                          |
| <b>Document Builder Module</b>      | Powers the interactive Resume and CV builder with downloadable PDF output.                               |
| <b>Collaboration Module</b>         | Provides real-time messenger chat and WhatsApp directly.                                                 |
| <b>Todo Module</b>                  | Task management system with CRUD operations stored in MongoDB.                                           |
| <b>Dashboard &amp; Admin Module</b> | Displays usage analytics, manages users, and tracks file downloads for administrators.                   |

## 8. Expected Outcomes

- A fully functional, deployed web application accessible from any modern browser.
- A complete PDF management suite covering merge, split, edit, convert, watermark, and signature tools.
- Important tools for speech-to-text, quiz generation, and PDF-to-audiobook conversion.
- Secure digital payment processing with subscription and billing management.
- An intuitive admin dashboard with real-time insights and user management.
- A built-in messenger or WhatsApp for real-time team collaboration within the platform.
- Professional resume and CV builders with PDF export capability.
- Improved productivity for students, professionals, and organizations by eliminating tool-switching.

## 9. Project Timeline (Tentative)

| Phase / Task                         | Duration                  |
|--------------------------------------|---------------------------|
| Requirement Analysis & Planning      | 1 week                    |
| UI/UX Design (Figma Prototypes)      | 2 weeks                   |
| Project Setup & Authentication       | 1 week                    |
| PDF Core Module Development          | 2 weeks                   |
| Conversion & AI Features             | 2 weeks                   |
| Payment Gateway Integration          | 1 week                    |
| Resume/CV Builder & Quiz Generator   | 2 weeks                   |
| Messenger/WhatsApp, Todo & Dashboard | 2 weeks                   |
| Admin Panel Development              | 1 week                    |
| Testing & Bug Fixing                 | 2 weeks                   |
| Deployment & Documentation           | 1 week                    |
| Total Estimated Duration             | ~17 weeks (Approximately) |

## 10. Conclusion

WeavePDF is proposed as a modern, full-featured, and user-centered SaaS platform designed to consolidate the fragmented world of digital document management and productivity tools. By combining PDF processing, AI capabilities, real-time collaboration, secure payments, and personal productivity features into a single, beautifully designed application, WeavePDF aims to save users time, enhance security, and deliver measurable improvements in daily workflow efficiency.

Built on a robust and industry-standard technology stack including Next.js, React, Node.js, MongoDB, JWT, and Tailwind CSS the platform is designed to be scalable, maintainable, and extensible. The first release will focus on delivering the core PDF and productivity features, with a clear roadmap for AI enhancements, mobile support, and enterprise-level capabilities in future versions.

WeavePDF has strong academic value as a comprehensive full-stack engineering project and significant real-world commercial potential as an independent SaaS product serving students, businesses, and professionals globally.